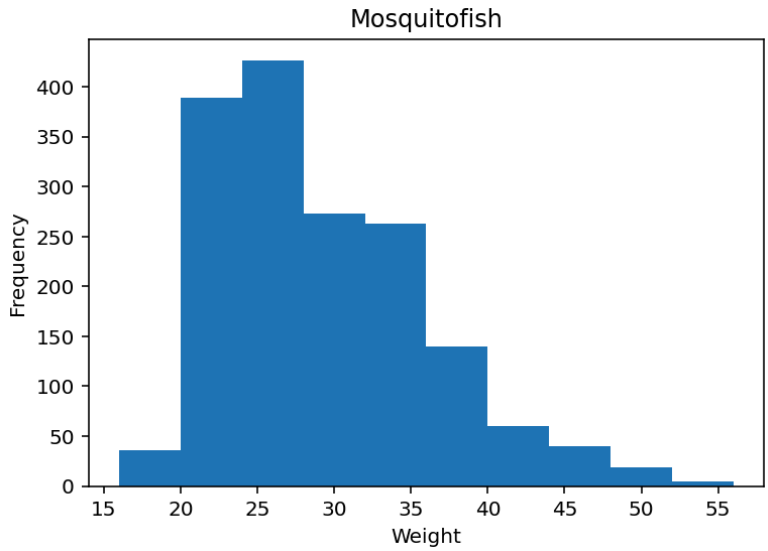
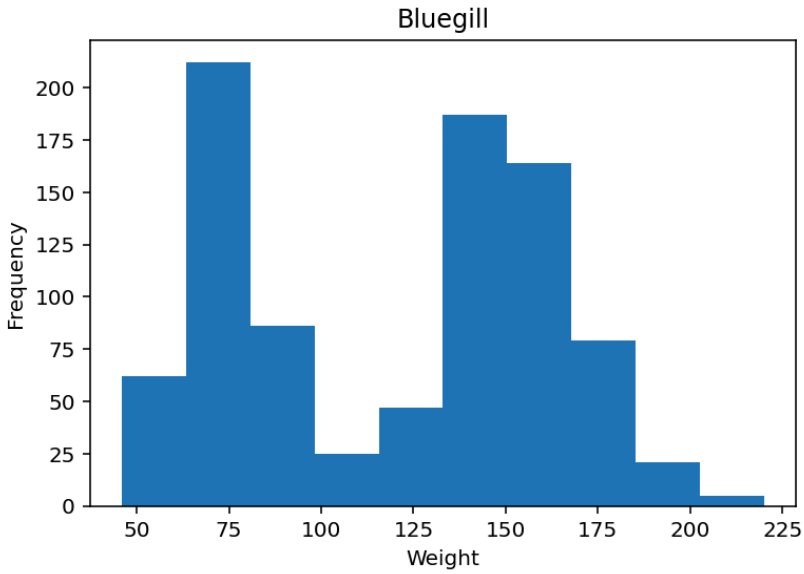
Hypothesis Testing with Z-scores and the Single-sample T-test

1) Plotting the data

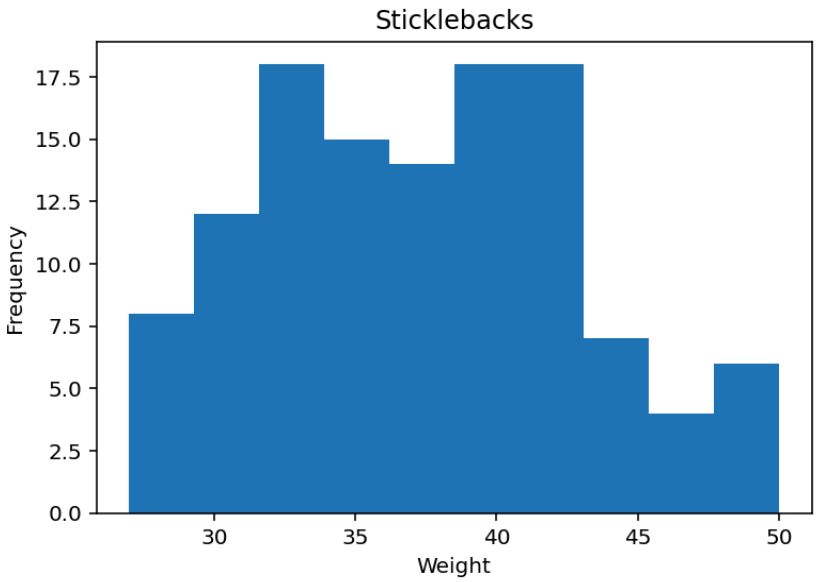
Histograms:
Mosquitofish Histogram:



Bluegill Histogram:



Sticklebacks Histogram:



1.2: Mean and standard deviation and standard deviation and display on the histogram:

Mosquitofish:

Mean: 120.03378378378379

Std Dev: 41.92484409451798

Bluegill:

Mean: 28.76075105996366

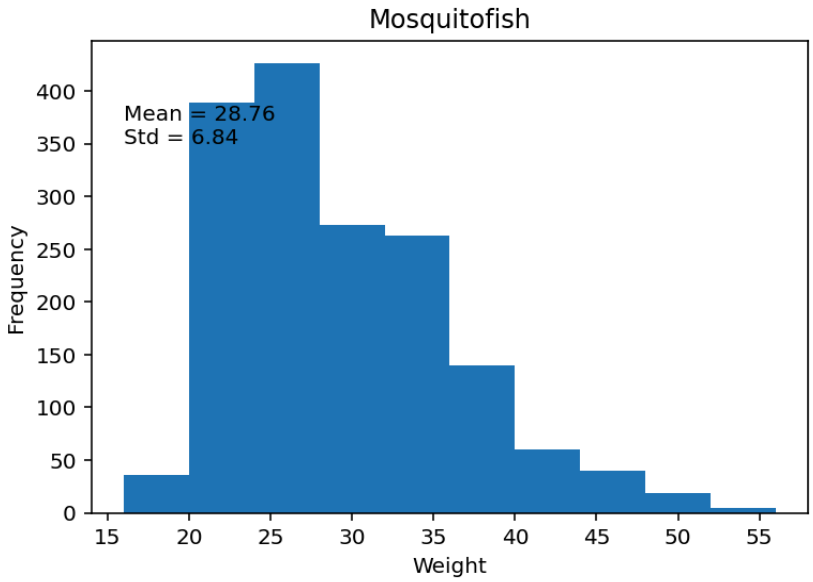
Std Dev: 6.835281714509543

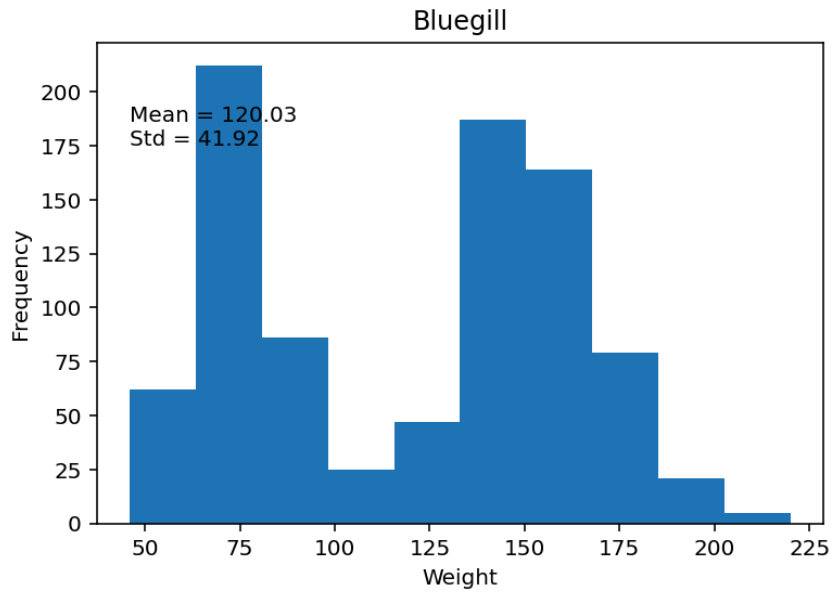
Sticklebecks:

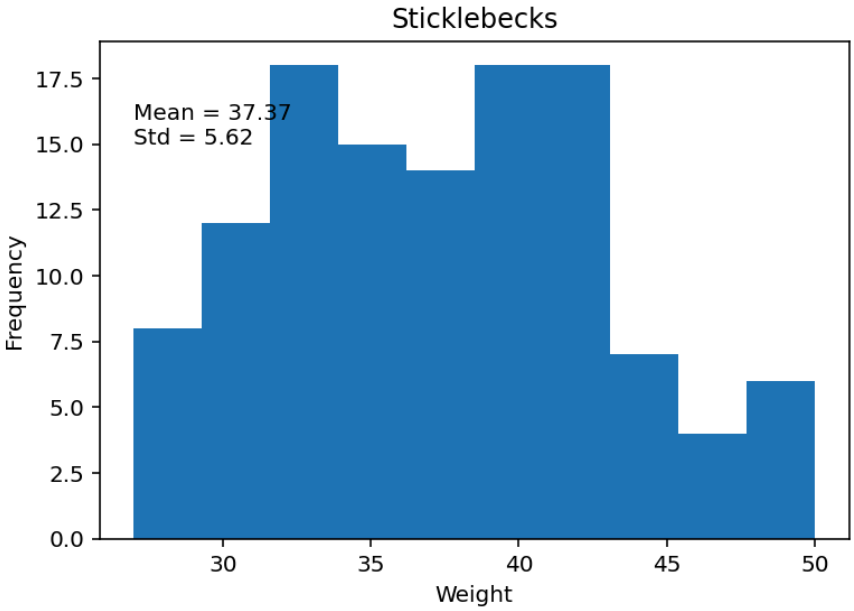
Mean: 37.36666666666667

Std Dev: 5.6212888992551235

Displayed on the histograms:







The three fish are pretty different in size. Bluegill are way bigger than the other two, with a mean of about 120mm and a huge standard deviation of 42, meaning there's a lot of size variation within that species. Mosquitofish and Sticklebacks are both small fish. Mosquitofish average around 29mm and Sticklebacks around 37mm, so they're closer to each other. Both of those have small standard deviations (~6-7) meaning their sizes are pretty consistent. The Mosquitofish histogram looks the smoothest because it has the most data points. Sticklebacks only has 120 fish so the histogram is a bit rougher looking.

2) Z-score for sample mean hypothesis test

2.1.1 Null and research hypotheses:

We're trying to figure out if the size of a sampled fish is significantly different from the Mosquitofish population, which would make size a useful feature for telling fish apart. For all three comparisons the hypotheses are the same:

H_0: The sample mean equals the Mosquitofish population mean (no difference)
H_1: The sample mean does not equal the Mosquitofish population mean (two-tailed)

2.1.2 Critical regions/values:
Two-tailed test so we reject H_0 if z falls outside:
α = 0.05: z < -1.96 or z > 1.96
α = 0.01: z < -2.576 or z > 2.576

2.1.3 Z-statistics and p-values:
Mosquitofish sample: z = -1.36, p = 0.1735
Bluegill sample: z = 30.76, p ≈ 0.000
Sticklebacks sample: z = 2.96, p = 0.0031

2.1.4 Effect sizes (where significant):
Bluegill: Cohen's d = 13.76
Sticklebacks: Cohen's d = 1.32
Mosquitofish: not significant, so no effect size calculated

2.1.5 Which tests were significant?
Bluegill and Sticklebacks were both significant at α = 0.05 and α = 0.01. Mosquitofish was not significant at either level.

3) Sample t-test
3.1.1 T-test results with n=5:
Mosquitofish: t = -1.69, p = 0.1663 — not significant at either level
Bluegill: t = 5.43, p = 0.0056 — significant at both α = 0.05 and α = 0.01
Sticklebacks: t = 2.88, p = 0.0449 — significant at α = 0.05 only

3.1.2 Comparison to z-test:
They're pretty similar but not identical. The biggest difference is Sticklebacks, the z-test gave p = 0.003 (significant at both levels) but the t-test gives p = 0.045 (only significant at 0.05).

3.1.3 Are 5 samples enough?
For Bluegill, yes I think the difference is so large that even 5 samples picks it up with both tests. For Sticklebacks, 5 samples is unsure. The z-test says yes at both levels but the t-test only gets α = 0.05. For Mosquitofish, 5 samples correctly shows no significance. So 5 samples is probably not enough to be confident.

3.1.4 Why t-test instead of z-test?

The z-test requires knowing the true population standard deviation of the fish you're testing. We only know the Mosquitofish population std dev. For Bluegill and Sticklebacks we don't have that.

3.2.1 T-test results with n=20:
Mosquitofish: t = 0.82, p = 0.420 — not significant
Bluegill: t = 13.02, p ≈ 0.000 — significant at both levels
Sticklebacks: t = 7.65, p ≈ 0.000 — significant at both levels

3.2.2 Comparison to n=5:
With 20 samples the results are a lot more clear cut. Sticklebacks went from barely significant to very significant.